Sectorization Using Genetic Algorithm

Models for Air Traffic Management

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Abstract: **This report aims to examine airspace sectorization using Genetic Algorithm (GA) combined with Voronoi diagrams. The relationship between ATC workload and the deployment of sectors within the Barcelona FIR using computational tools to address sectorization has been analyzed.**

Keywords: Air Traffic Control (ATC), Sectorization, Genetic Algorithm (GA), Voronoi diagrams, Pareto front.

I. Introduction

Air Traffic Control (ATC) is essential for ensuring a safe and efficient flow of air traffic in the context of current aviation, which faces an increasing air travel demand and, consequently, higher workload levels for controllers. It is fundamental to manage ATC workload distribution effectively, as it significantly influences their decision-making capacity and overall performance.

Airspace sectorization, the process devoted to dividing the airspace into smaller sectors, is a useful strategy for uniformly balancing workload distribution. Each sector is assigned to a specific ATC team while maintaining safety and operational efficiency.

To address airspace sectorization, this project uses Genetic Algorithm (GA) and Voronoi diagrams in the Barcelona FIR, which has been selected as the focus of this study.

II. Objective and definitions

To begin with, as it has been introduced in the previous section, the purpose of this study is to optimize airspace sectorization using Genetic Algorithm (GA) and Voronoi diagrams. The ATC workload distribution must be balanced within the Barcelona FIR.

ATC tasks, which can be divided into four categories, are defined below:

- Background tasks: routine activities such as system configuration, staff coordination and weather monitoring. They do not depend on traffic volume.

- Transition tasks: tasks related to the hand-off and initial management of aircraft between sectors, including establishing communication and planning the initial route.

- Recurring tasks: continuous activities such as traffic monitoring, restricted areas avoidance and maintaining separation between aircraft.

- Conflict tasks: actions focused on detecting and resolving conflicts appearing between aircraft, including vectoring and post-conflict route recovery.

To continue, Genetic Algorithm (GA) simulates natural selection and evolves optimal solutions over time, which is useful for designing an optimal airspace sectorization.

When it comes to Voronoi diagrams, it must be mentioned that it is a way of dividing the space into regions around a generating point. Moreover, every point within a region is closer to its generating point than to any other because the regions' boundaries are defined by perpendicular bisectors of the lines connecting the generating points and, therefore, are equidistant of the two cells they separate.

Real air traffic data from a day of July of 2008 in which all flights fly at FL300 or higher, provided by Eurocontrol, is used to sectorize the Barcelona FIR. It has been used a file containing data from a period of time of the day with high traffic: LatLon-FIR-FL300up-filt-0755-0810.dat.

Some constraints have been taken into consideration when performing the sectorization. First of all, no sector should contain more than 15 to 17 aircraft, which has been considered as the maximum number of flights which a controller can manage simultaneously. Secondly, the number of sectors needs to be minimized. Nevertheless, at least two must be deployed.

III. Single algorithm sectorization optimization

Firstly, the choice of the complexity parameters' values has been made to perform the single algorithm sectorization optimization.

It has to be mentioned that the population size (PS) and the maximum number of generations (MaxGen) values are the optimal ones, derived from the Monte Carlo analysis, detailed in *IV. Monte Carlo Optimization*. The rest of the parameters have been selected based on what has been learned in WP0.

CHOICE OF COMPLEXITY PARAMETERS:	
Maximum number of aircraft per sector	15 aircraft
Number of sectors (NSECT)	The ceiling of the result of dividing the total number of aircraft in the FIR by the maximum aircraft which a sector can contain
Number of variables (NVAR)	(NSECT x 2) as each sector is defined by a Voronoi point located at coordinates {x,y}

Lower bounds (LB)	Since Voronoi points can be located outside the FIR but not at an infinite distance, the lower and upper bounds for their coordinates are the same limits used by the FIR map.
Upper bounds (UB)	
Population size (PS)	50 individuals
Maximum number of generations (MaxGen)	20 generations
Crossover Fraction [0 - 1] (CF)	0,8
EliteCount (EC)	(0,1 x PS) individuals
Mutation fraction [0 - 1] (MF) (Only for AGA)	0,2
Newcomers fraction [0 - 1] (NF) (Only for AGA)	0,2

Table 1: Choice of complexity parameters.

It is important to note that the chosen number of maximum aircraft per sector is 15 because, even though the tasks performed by the controller need to be maximized within the established limits, it is wise to avoid the possible solutions which are very close to the operational limit in case something went wrong if those solutions were applied in real life.

Next, to define the objective function which will be optimized by the GA, the conflict points appearing in airways intersections have been considered. It has been decided that they are included in the ATC conflict tasks, as conflicts arise and must be detected and resolved, which is the primary objective of this task category.

The objective function, which is minimized by the GA, is composed by the relative standard deviation (RSD) of the airways intersections, considering that the controllers' workload is defined by them. Minimizing the RSD, we ensure a balanced distribution of workload in all the sectors because the dispersion of values relative to its mean is minimum.

The RSD is utilized in order to have the values normalized for future analysis in which more than one objective function will be considered.

It has to be mentioned that, due to the inherent behaviour of the GA, the results obtained change every time it is executed. Nevertheless, as they tend to have similar values, the runs performed with optimal values for PS and MaxGen parameters have been included in this report so as to interpret the results.

A. AGA

Firstly, the AGA has been executed. The numerical results obtained and their interpretation are detailed in *VI. Analysis of results*.

The airspace sectorization is presented below:

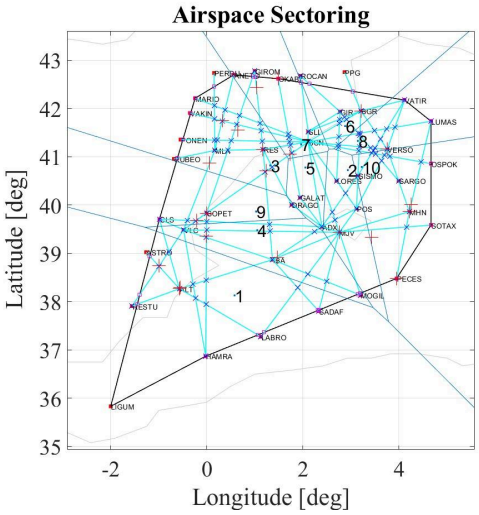


Figure 1: Airspace sectorization obtained with AGA.

As it can be observed, 10 sectors have resulted from the execution of the algorithm.

B. GA (Matlab)

Secondly, the GA (Matlab) has been executed. The numerical results obtained and their interpretation are detailed in *VI. Analysis of results*.

The airspace sectorization is the following one:

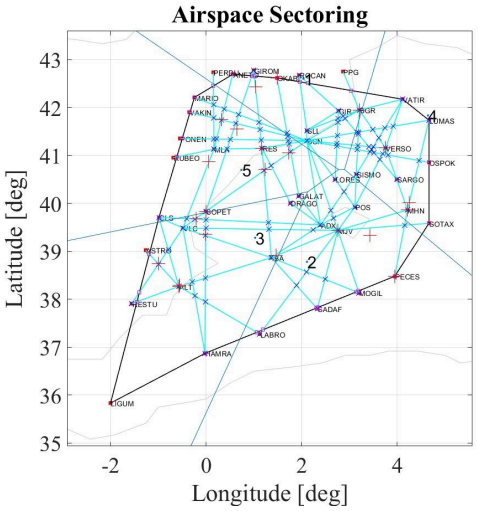


Figure 2: Airspace sectorization obtained with GA (Matlab).

In this case, 5 sectors have resulted from the execution of the algorithm.

In addition, the following plots are obtained:

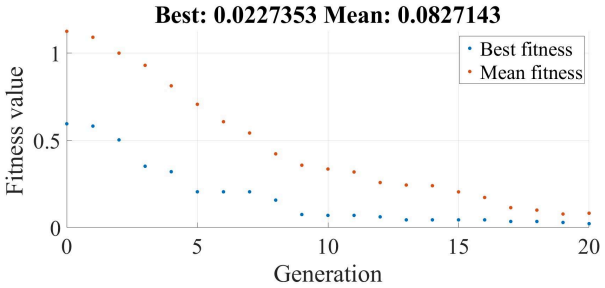


Figure 3: Evolution of best and mean fitness values with GA (Matlab).

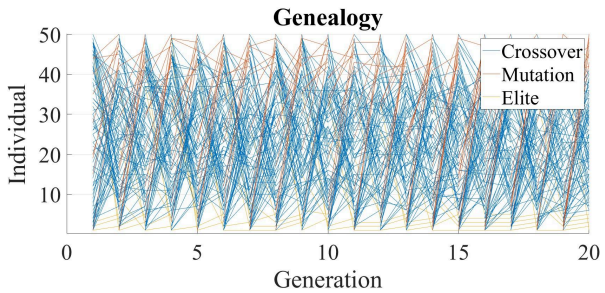


Figure 4: Genealogy with GA (Matlab).

In Figure 3, it can be observed that the more generations appear, the smaller the best and mean fitness values tend to be. This behaviour demonstrates that, increasing the number of generations, it is more feasible to find the desired minimum value. With regard to Figure 4, it shows the evolution of individuals along several generations.

Moreover, find attached in the *Annexes* the numerical results obtained for all the complexity parameters for each sector using both the AGA and the GA (Matlab).

Finally, find in the *Annexes* the sectorization obtained with non-optimal values (initial setup values): PS = 100 and MaxGen = 30. It can be observed that the cost for both algorithms is generally higher than with optimal values while obtaining the same amount of sectors deployed in the FIR. As it happens in the optimal scenario, the GA (Matlab) offers a smaller cost and amount of sectors than the AGA, which is aligned with the project's objective.

IV. Monte Carlo analysis

To continue, to analyze the influence of the population size (PS) and the maximum number of generations (MaxGen), a Monte Carlo analysis is performed.

It consists of executing, in this case, the GA (Matlab) several times for a set of parameters and plotting the results obtained. When the cumulative mean of the results converges, a maximum number of iterations (MC) can be derived from the plot. MC is used later on to obtain the optimal values for PS and MaxGen.

The values used for PS and MaxGen to perform the Monte Carlo analysis of convergence are different to the ones used in the previous section. These were the initial setup values used to determine the number of iterations required to find the optimal ones used in *III. Single algorithm sectorization optimization*: PS = 100 and MaxGen = 30.

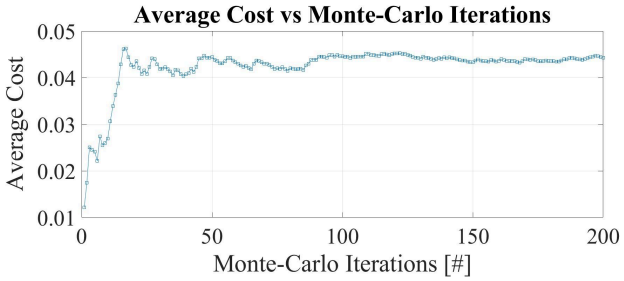


Figure 5: Average cost vs Monte Carlo iterations.

The value selected for MC has been 80 iterations, which is when the cumulative mean first converges for a reasonable number of iterations.

After that, the algorithm results have been plotted for several combinations of PS and MaxGen values to obtain the optimal ones.

VALUES FOR THE MONTE CARLO ANALYSIS	
PS values	[10, 20, 30, 50, 70] individuals
MaxGen values	[10, 20, 30, 50, 70] generations
Iterations (MC)	80 iterations
Total runs	2000 runs

Table 2: Values for performing the Monte Carlo analysis.

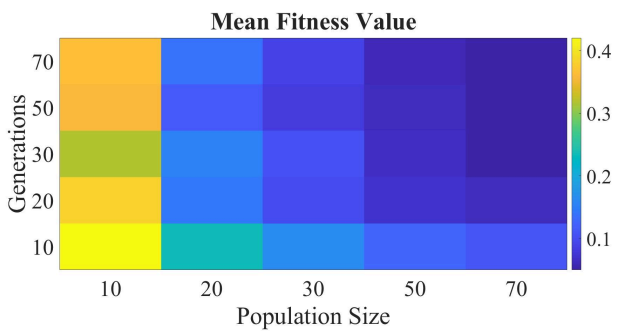


Figure 6: Mean fitness value.

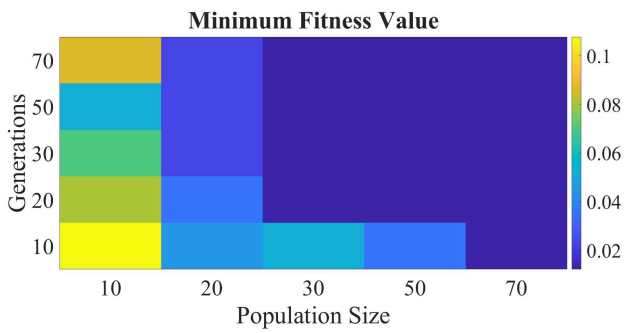


Figure 7: Minimum fitness value.

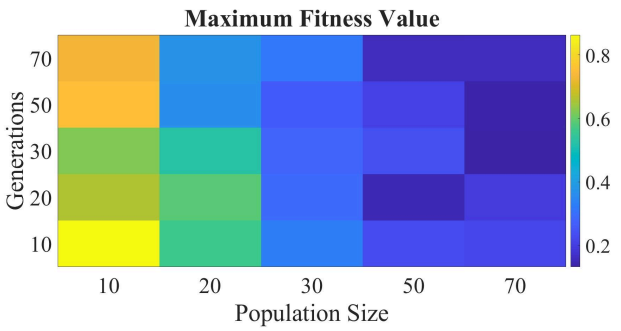


Figure 8: Maximum fitness value.

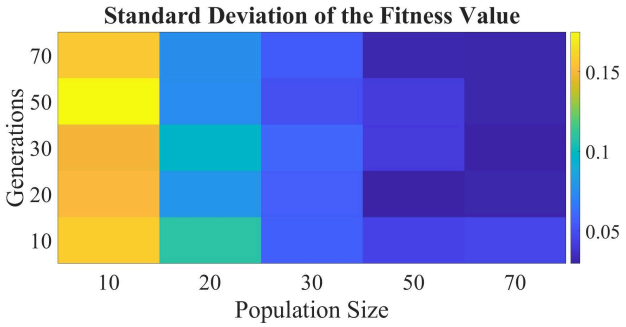


Figure 9: Standard deviation of the fitness value.

Using Figure 6, it has been possible to determine a range of combinations in which the mean takes the smallest values: starting with PS = 30 and MaxGen = 20 and increasing both values. Furthermore, Figure 7 has demonstrated that these combinations also achieve the smallest values. Moreover, with Figure 8, the collection of combinations can be reduced to select those for which the maximum values are also minimum. It happens for PS = 50 or larger. Finally, Figure 9 confirms that the set obtained at this point (starting with PS = 50 and MaxGen = 20 and increasing both parameters) also has the minimum values for the standard deviation, showing little dispersion.

To conclude, according to the reasons which have just been stated and taking into consideration that the minimum values need to be obtained, it has been decided that the optimal parameters are the following ones:

Population size (PS)	50 individuals
Maximum number of generations (MaxGen)	20 generations

Table 3: PS and MaxGen derived from the Monte Carlo analysis.

V. Multi-objective GA sectorization optimization

To start with, it is important to realize that the number of aircraft per sector which appear when using the single-objective algorithm does not always fulfill the first constraint.

From now on, two different objectives will be optimized: firstly, the airways intersections and, subsequently, the conflict points in each sector. Secondly, the number of aircraft within each region, which has been considered as a recurring task due to the fact that it can be associated with the traffic monitoring process done continuously.

When optimizing both objectives, a set of equally optimal solutions will be obtained, known as Pareto front. A trade-off between objectives is presented within each solution. Another important aspect must be considered, the rank, which is a way of categorizing the individuals. The ones with the best fitness value, have the lowest rank. Therefore, the solutions which are not dominated by any other are the ones with minimum rank. The Pareto fraction which has been used is PF = 0,5.

Using the set of optimal parameters for the single-objective algorithm, a non-optimal result is obtained with multi-objective optimization. See the results in the *Annexes*. It is a non-optimal result because, if the analysis is based on the Pareto and Rank histograms, the objective is to have the biggest amount of solutions in Rank 1 and significantly less in the following ranks. In this case, 12 solutions appear in Rank 1 and 9 in Rank 2, which shows that the difference between them is not large enough to consider it an optimal solution.

Nevertheless, analysing the possible combinations of PS and MaxGen parameters, it has been concluded that the bigger the population size and, especially, the larger the number of maximum generations, the more individuals are placed in Rank 1 and more difference appears between the not dominated solutions and the ones in Rank 2 and lower.

Subsequently, the following selection of PS and MaxGen parameters' values has been performed:

Population size (PS)	70 individuals
Maximum number of generations (MaxGen)	50 generations

Table 4: PS and MaxGen for the multi-objective optimization.

With this set of parameters, the influence of the Pareto fraction has been studied for different values: [0.1, 0.5, 0.9].

It can be derived that the optimal value for PF is the one presented below. With it, the biggest difference in the number of individuals between the first and the second rank can be achieved: 42 in Rank 1 and 12 in Rank 2.

Pareto fraction [0 - 1] (PF)	0,6

Table 5: Selection of the optimal Pareto fraction (PF) parameter.

Observe below the Pareto front and the Rank histogram, as well as the sectorization obtained for the first solution:

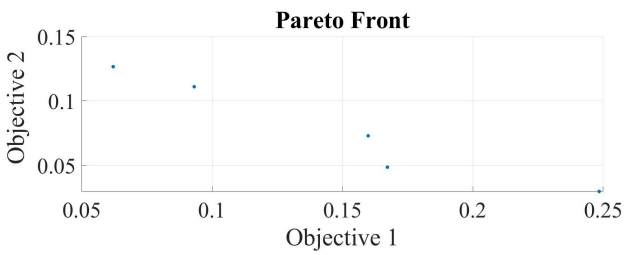


Figure 10: Pareto front.

It is important to note that, in Figure 10, only the first Pareto front is plotted, the one containing the Rank 1 solutions.

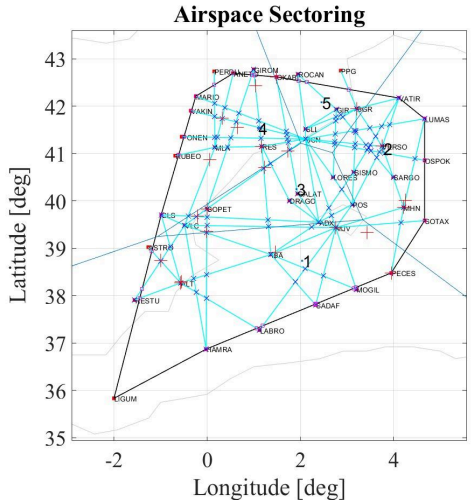


Figure 11: Airspace sectorization obtained with the first solution.

In this case, 5 sectors have been obtained.

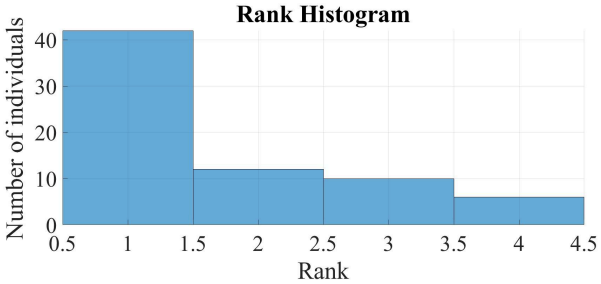


Figure 12: Rank histogram.

As it can be seen in Figure 12, most of the solutions are concentrated in Rank 1.

The numerical results obtained and their interpretation for the first individual in the Pareto front are detailed in *VI. Analysis of results*. Furthermore, two more plots with additional information are attached in the *Annexes*, as well as the numerical results for all the complexity parameters' values.

VI. Analysis of results

	SINGLE OBJECTIVE		MULTI-OBJECTIVE (MOO)
	AGA	**GA (Matlab)**	**GA multi-objective**
Objective function			
Best cost / Cost first indiv. for MOO	0,29	0,02	0,16 0,07
Best individual / First individual in the Pareto front for MOO	0,5767 38,1299 2,9344 40,7234 1,3212 40,8183 1,0487 39,4770 2,0522 40,7768 2,8794 41,6337 1,9645 41,2532 3,1439 41,3275 1,0267 39,8667 3,2244 40,7877	2,0388 42,6014 2,1129 38,7816 1,0276 39,2824 4,6450 41,8344 0,7402 40,6975	2,0373 38,7368 3,7738 41,1038 1,9159 40,2485 1,0904 41,5377 2,4551 42,0826
Relevant complexity parameters			
Airways intersections	16, 21, 25, 22, 9, 17, 23, 22, 11, 18	37, 37, 36, 38, 36	38, 39, 39, 33, 35
Aircraft in sector	3, 5, 15, 5, 0, 2, 13, 2, 10, 17	13, 7, 13, 17, 22	14, 17, 14, 16, 11

Table 6: Sectorization optimization numerical results.

To begin with, the single-objective algorithms present different behaviours. On the one hand, the AGA provides a sectorization with 10 airspace sectors and a best cost of 0,29. On the other hand, the GA (Matlab) is able to sectorize the airspace in only 5 sectors and with a best cost of 0,02, nearly 10 times smaller. Furthermore, the distribution of workload is far more balanced in the latter case, as values are closer to the mean than with the AGA.

Having said that, it has been demonstrated that it is more convenient to execute the GA (Matlab) than the AGA when sectorization is performed with a single objective.

To continue, it might be necessary to optimize more than one objective, as it has happened in the last phase of this project. The first solution of the GA multi-objective presents a sectorization composed of 5 airspace sectors and relatively low costs, especially compared to the AGA. Moreover, there is very little dispersion in the complexity parameters, which demonstrates an optimal balance of workload. Therefore, it can be said that this solution obtained with MOO is robust enough to perform an optimal sectorization.

VII. Conclusions

Once the results have been analyzed, it can be concluded that the GA (Matlab) is the most optimal algorithm to perform a single-objective sectorization.

When it comes to multi-objective sectorization, the GA multi-objective has demonstrated that it represents a very good tool to perform it.

A. Numerical results for all the complexity parameters obtained with AGA:

Complexity:

aircraftInSector	firTransfers	sectorTransfers	airwaysIntersections	airwaysLength	sectorArea
3	9	8	16	1634.3	73363
5	0	16	21	997.54	18951
15	5	13	25	1599.6	30871
5	2	22	22	1569.3	31567
0	0	14	9	551.66	9402.8
2	2	15	17	734.62	11608
13	5	19	23	1060	16864
2	5	16	22	842.35	12655
10	3	13	11	659.8	22965
17	3	12	18	1095.5	31349

Figure 13: Complexity parameters obtained with AGA.

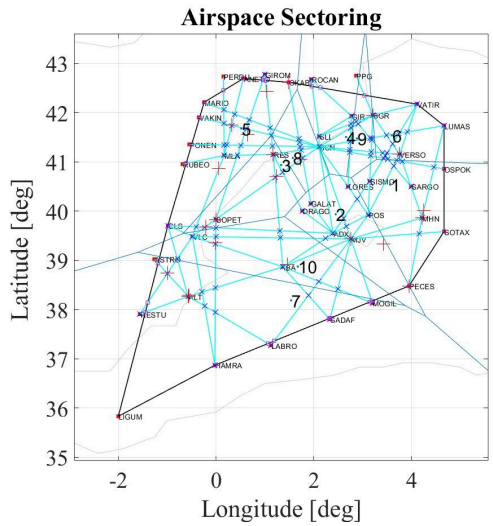
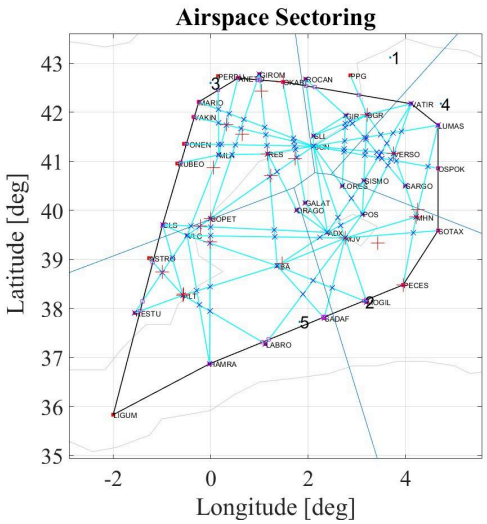
B. Numerical results for all the complexity parameters obtained with GA (Matlab):

Complexity:

aircraftInSector	firTransfers	sectorTransfers	airwaysIntersections	airwaysLength	sectorArea
13	7	21	37	1700.2	28808
7	7	19	37	2506.3	59721
13	5	18	36	2448.8	84946
17	7	13	38	1620	32359
22	8	27	36	2469.5	53624

Figure 14: Complexity parameters obtained with GA (Matlab).

C. Single-objective sectorization obtained with non-optimal values (PS = 100 and MaxGen = 30):

AGA	***GA (MATLAB)***
	

Objective function			
Best cost	Best individual	Best cost	Best individual
0,36	3,5613 40,5388 2,4498 39,9369 1,3319 40,9354 2,6620 41,5042 0,5145 41,6776 3,5997 41,5361 1,5356 38,1870 1,5706 41,0811 2,8849 41,4760 1,6715 38,8736	0,06	3,6876 43,1154 3,1594 38,1413 -0,0016 42,5962 4,7252 42,1755 1,8254 37,7277

Complexity parameters

AGA

Complexity:

aircraftInSector	firTransfers	sectorTransfers	airwaysIntersections	airwaysLength	sectorArea
15	2	13	16	1081.2	28608
7	1	19	32	1370.7	25896
20	0	17	15	785.61	20483
0	2	15	14	808.25	12777
16	12	11	21	1534.5	34472
4	5	14	24	788.48	14725
3	11	10	16	1546.2	69762
2	0	15	18	722.22	10956
0	1	18	7	518.41	6253.9
5	0	24	21	1589.3	35549

GA (MATLAB)

Complexity:

aircraftInSector	firTransfers	sectorTransfers	airwaysIntersections	airwaysLength	sectorArea
2	3	16	35	1361.3	18052
22	4	18	38	2158.2	48779
33	12	14	39	2691.2	59774
2	6	16	34	1490.4	28231
13	9	16	38	3043.7	1.0454e+05

Table 7: Sectorization with single-objective optimization with non-optimal values.

Additional plots obtained with GA (Matlab) with non-optimal values:

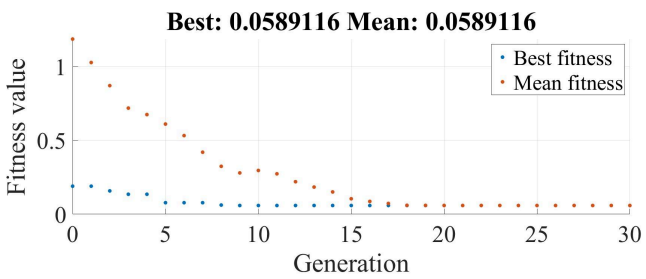


Figure 15: Evolution of best and mean fitness values with GA (Matlab) with non-optimal values.

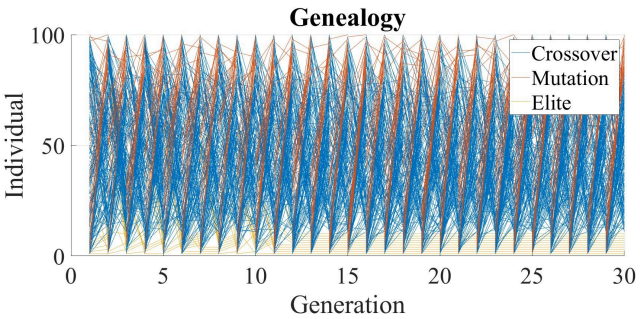
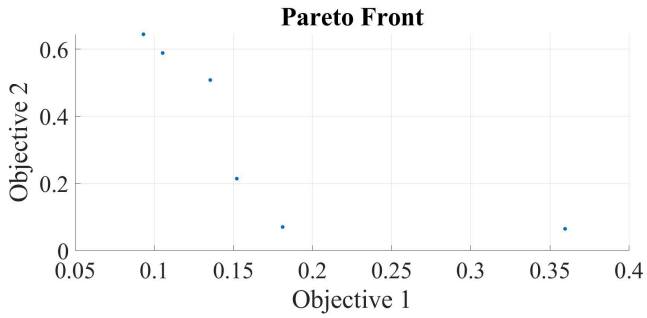
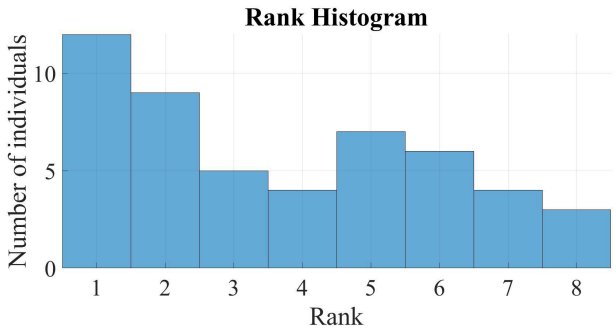


Figure 16: Genealogy with GA (Matlab) with non-optimal values.

D. Multi-objective optimization obtained with non-optimal values (PS = 50 and MaxGen = 20):

GA MULTI-OBJECTIVE





Objective function	
Cost of the first individual in the Pareto front	First individual in the Pareto front
0,15 0,21	0,9937 40,6800 3,7731 41,0795 2,0247 39,5014 2,3579 41,4035 0,3930 38,6190

Complexity parameters

Complexity:

aircraftInSector	firTransfers	sectorTransfers	airwaysIntersections	airwaysLength	sectorArea
15	6	16	25	2088.4	50371
17	7	13	39	1941.1	40893
11	5	17	43	2442.7	51877
15	6	18	44	2012.9	32017
14	10	8	33	2259.7	83615

Table 8: Multi-objective optimization with non-optimal values.

E. Numerical results for all the complexity parameters obtained with GA multi-objective with optimal values:

Complexity:

aircraftInSector	firTransfers	sectorTransfers	airwaysIntersections	airwaysLength	sectorArea
14	12	14	38	3126.1	1.1025e+05
17	7	15	39	1965.8	41798
14	0	29	39	2185.1	39554
16	11	15	33	2292.5	49965
11	4	17	35	1175.2	17654

Figure 17: Complexity parameters obtained with GA multi-objective.

F. Additional plots obtained with GA multi-objective with optimal values:

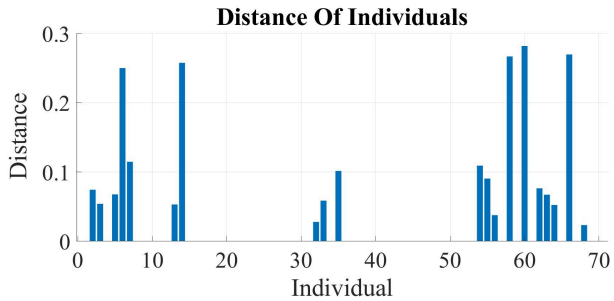


Figure 18: Distance of individuals.

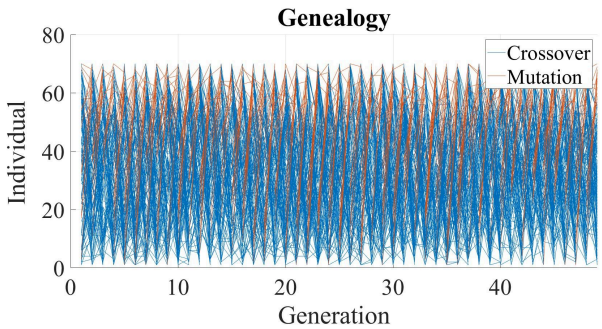


Figure 19: Genealogy.

G. Contributions:

All the members of the group have contributed equally to the completion of this project.